2026

Architecture Decision Log

HEALTHCARE ANALYTICS CLOUD FOUNDATION

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DOCUMENT REVISION LOG

REV.	UPDATE DATE	BY
CREATED	JULY 8, 2026	ALEJANDRA
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ID	Decision	Rationale	Alternatives Considered	Impact	Sprint	Status
ED-001	Use a single Resource Group for the development environment	Grouping resources based on lifecycle simplifies resource organization and supports lifecycle management	Multiple Resource Groups by service; separate Resource Groups by functional area; unmanaged resource organization	Reduces administrative overhead and simplifies cleanup	Sprint 0	Approved
ED-002	Implement a reference development environment to validate the solution architecture before future production deployment	Supports iterative implementation while minimizing operational cost and risk	Production-only deployment; simultaneous Development and Production environments; local-only development without Azure resources	Enables rapid development while controlling cloud expenditure	Sprint 0	Approved
ED-003	Use GitHub OIDC Federated Credentials for Azure authentication	Eliminates long-lived secrets and aligns with modern cloud security practices	Client Secret; Certificate	Improves security, reduces credential management overhead, supports passwordless automation	Sprint 1	Approved
ED-004	Portal-First Design with Reverse-Engineering IaC Blueprinting	For the purpose of this portfolio project the decision was made to use Azure Portal first design vs. IaC design	Hand-writing Bicep from scratch	Ensured building the cloud foundation step by step while reviewing key concepts as part of the learning process of this cloud foundation portfolio project	Sprint 2	Approved

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ED-005	Deploy the Core Network Foundation (VNet) prior to provisioning the Analytics Data Storage Account	Aligns with the Azure Cloud Adoption Framework's security-first design	Deploying storage first	Security-first design with a private network available when creating storage assets to prevent any temporary public exposure of the storage account endpoints	Sprint 2	Approved
ED-006	Leave Virtual Network Encryption disabled	The solution does not include VM-to-VM traffic requiring VNet encryption. Azure Storage provides encryption at rest, and Private Endpoints with TLS secure data in transit.	Enable VNet Encryption	Avoids unnecessary complexity while maintaining an appropriate security posture for the development environment.	Sprint 2	Approved
ED-007	Architectural Transition to a Serverless PaaS/SaaS Lakehouse Engine with Future Microsoft Fabric integration	Minimizes security compliance overhead, eliminates VM cost, removes OS patching, and directly aligns with modern cloud analytics practices; Eliminates the operational overhead and memory constraints of traditional Power BI import refreshes	Traditional Power BI scheduled import refreshes or high-latency DirectQuery SQL views with Virtual Machine (VM) resource for compute power	Drastically drops platform support complexity; Eliminates VM cost; Physical storage footprints scale dynamically on-demand, compute fees drop to zero when inactive, and data duplication bottlenecks are entirely bypassed	Sprint 2	Approved

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ED-008	Configure data storage using the Standard Performance tier and LRS replication	Standard performance meets baseline data lake batch ingestion throughput requirements at the lowest cost point.	Premium Performance tiers; Geo-Redundant Storage (GRS) replication.	Minimizes subscription credit consumption while keeping data safe within a single datacenter zone.	Sprint 2	Approved